ABHINAV GUPTA

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EDUCATION

University of Waterloo Sep 2020 - May 2025
Bachelor of Honours Computer Science with Distinction – Software Engineering Specialization. **3.7 GPA**
Relevant Courses – Intro to AI, Algorithms, Data Structures, User Interfaces, Database Management, Software Design

WORK EXPERIENCE

Software Development Engineer I June 2025 – Present
Amazon - Alexa *Toronto, ON*

- Launched GenAI-powered personalization systems for **Alexa** devices in **Java**, generating context-aware and timely recommendations from user signals (e.g., reminders, weather, conversation history), increasing customer engagement by **20%**.
- Engineered a real-time event-driven pipeline for live sports updates (e.g., NFL halftime events), generating **500K+ notifications** using LLMs per day, increasing customer engagement by **40%**; improved latency through concurrent processing;
- Optimized LLM latency, operational costs, and **AWS Bedrock** inferencing capacity for **Anthropic** models by up to **70%** by optimizing prompts, introducing dynamic prompt generation, and caching input tokens across production pipelines.
- Engineered an **LLM-as-a-Judge** layer to evaluate AI-generated personalized podcast topics for **Alexa** using a pipeline architecture, automatically invalidating **30%** of low-quality topics, processing **10,000** beta customers daily.
- Built and owned a business metrics and observability pipeline (**AWS CloudWatch, S3, Lambda, SQS**) to track user engagement, system performance, and LLM performance, enabling data-driven decision-making for product launches.
- Drove a **90% reduction in data compliance risks** through privacy and security reviews for GenAI services; enforced **GDPR** (DSAR, OD3) and **DMA** compliance, implemented safeguards against prompt injections, sanitized LLM inputs/outputs, and integrated real-time **PII detection** on LLM inferencing with **AWS Bedrock Guardrails**.

Machine Learning Engineer Intern May 2024 – Dec 2024
Bell *Toronto, ON*

- Built an automated service with **Gemini** LLM to summarize technician work, cutting tech support response times by **70%**.
- Executed DevOps to deploy custom-trained ML models stored as **Docker** images using **Terraform** and **GCP**, enhancing deployment efficiency by **80%**; deployed models used by Field Technicians across Canada.
- Collaborated with Data Engineers to develop a **GitLab CI/CD** pipeline for deploying **GCP CloudSQL** schema changes with **AtlasGo**, **Bash**, and **Makefile**, eliminating DDL mismatches and reducing data migration errors by **90%**.
- Developed a robust **BentoML** unit testing framework with **Python**, **Pytest**, and **Docker** to validate model and API logic for AI projects used by Bell's Data Scientists, reducing errors in production by **50%**.

Cloud Backend Engineer Intern May 2023 – Aug 2023
Deloitte *Toronto, ON*

- Engineered a system to enable users to search for data from 2,000+ cloud bundles stored in **MongoDB** using **C#** and **.NET**, implementing advanced string matching and a dynamic Mongo query builder for streamlined data retrieval.
- Built a backend service that schedules task execution for cloud account provisioning - accelerating the process by **50%**.
- Constructed an object-oriented caching service using **Redis Cache** to enhance cloud account data retrieval speeds by **10x**.

Backend Engineer Intern Jan 2023 – Apr 2023
OpenLane *Toronto, ON*

- Implemented **AWS Serverless** REST APIs in **TypeScript** with **PostgreSQL** and **Elastic Search** databases to execute business logic for online car auctioning, reducing operational costs by **30%** compared to traditional server hosting.
- Built a **NodeJS** script to run multiple threads in parallel to retrieve 6 million vehicle data from **OracleDB** to update development databases weekly. Improved performance by **160%** when compared to single-threaded querying.
- Formulated 40+ **Cucumber** integration, **Jest** unit, and **Mocha** unit test cases - increasing code reliability by **15%**.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, HTML/CSS
Frameworks: AWS, GCP, React, Node, FastAPI, LangChain, .NET, Express, TailwindCSS, NextJS, Swagger, Cucumber, Jest
Developer Tools: Git, Linux, Docker, TerraForm, Kubernetes, Redux, OpenShift, MongoDB, PostgreSQL, OracleDB, Elastic Search, Postman, DBeaver, Studio3T, Redis, Heroku, Jenkins